1. $\log_x(y^x) = 10$

$$x \log_x(y) = 10$$

Using the change of base formula, $\log_x(y) = \frac{\ln y}{\ln x}$. Since $\log_y x = \frac{9}{4}$, we have $\log_x y = \frac{4}{9}$. Thus,

$$x \cdot \frac{4}{9} = 10 \implies x = \frac{10 \cdot 9}{4} = \frac{90}{4} = \frac{45}{2}$$

2. $\log_y(x^4 y) = 10$

$$4 \log_y x + 1 = 10 \implies 4 \log_y x = 9 \implies \log_y x = \frac{9}{4}$$

Again, using the change of base formula, $\log_x y = \frac{4}{9}$.

Next, we express x and y in terms of each other:

$$x = \frac{45}{2} \quad \text{and} \quad y = \left(\frac{45}{2}\right)^{4/9}$$

We then compute the product xy:

$$xy = \left(\frac{45}{2}\right) \cdot \left(\frac{45}{2}\right)^{4/9} = \left(\frac{45}{2}\right)^{1+4/9} = \left(\frac{45}{2}\right)^{13/9}$$

However, upon re-evaluating and using another approach by setting $x = 2^a$ and $y = 2^b$, we find:

$$\log_x(y^x) = 10 \implies \frac{x \log_x y}{\log_x y} = 10 \implies x \cdot \frac{4}{9} = 10 \implies x = \frac{45}{2}$$

$$\log_y(x^4 y) = 10 \implies \frac{4 \log_y x + 1}{\log_y x} = 10 \implies 4 \cdot \frac{9}{4} + 1 = 10 \implies 9 + 1 = 10 \quad \text{consistent}$$

Finally, we find that the product xy simplifies to:

$$xy = \left(\frac{45}{2}\right)$$